

SHWETA JAIN

Artificial Intelligence, Machine Learning & Quantum Researcher

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EDUCATION

National Institute of Technology Delhi, India

Department of Applied Sciences

Master of Technology (M. Tech) in Mathematics & Computing

GPA: 9.8/10 (Up to 1st Year) – Rank1

Aug. 2024 - Present

Kurukshetra University, Haryana, India

Panipat Institute of Engineering & Technology, Panipat

Department of Artificial Intelligence & Machine Learning

Bachelor of Technology in Computer Science & Engineering

Specialization in Artificial Intelligence & Machine Learning

GPA: 9.43/10 (Gold Medalist)

Sep.2020 - June 2024

C.C.A.S Jain Sr. Sec. School, Gannaur, Sonipat, Haryana, India

Higher Secondary Examination (CBSE Board)

Aggregate: 95.0% (CBSE School Branch Topper)

Apr. 2019 - Mar 2020

C.C.A.S Jain Sr. Sec. School, Gannaur, Sonipat, Haryana, India

Secondary Examination (CBSE Board)

Aggregate: 89.3%

Apr. 2017 - Mar 2018

RESEARCH INTEREST

Artificial Intelligence & Machine Learning, Deep Learning, Deep Reinforcement Learning, Computer Vision, Natural Language Processing, Image Processing, Quantum Computing & Quantum Machine Learning, Optimizations, Large Language Modelling

TECHNICAL SKILLS

- **Languages** Python, C/C++, HTML, CSS, SQL, R, JavaScript
- **Tools and Frameworks** GitHub, Visual Studio, Jupyter Lab & Notebook, R Studio, Google Collab, Arduino IDE, MATLAB
- **Machine Learning, Deep Learning & Generative AI** ML Libraries (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn) & Models, CNN, RNN, Time Series Forecasting, TensorFlow, Keras, Transformers, Hugging Face, LLMs, Deep Reinforcement Learning
- Quantum Computing & Quantum Machine Learning
- **Internet of Things** Arduino and Raspberry Pi

PUBLICATIONS

1. Aggarwal, Rishit, **Shweta Jain**, Anju Bhandari Gandhi, Drishti Dhingra, and S. C. Gupta. "Artificial Intelligence and Machine Learning in Women's Health: Unveiling PCOS with Image Analysis." In *Computational Intelligence and Mathematical Applications*, pp. 247-252. CRC Press, 2024. Doi: 10.1201/9781003534112-40.

2. Jain, Himanshi, **Shweta Jain**, Anju Bhandari Gandhi, Rishit Aggarwal, and S. C. Gupta. "Technological Marvels: Pioneering PCOS Detection through the Lens of Machine Learning and Deep Learning." In *Computational Intelligence and Mathematical Applications*, pp. 258-264. CRC Press, 2024. Doi: 10.1201/9781003534112.
3. **Jain, Shweta**, Anju Bhandari Gandhi, Stuti Mehla, Rishit Aggarwal, and Tanu Kwatra. "Virtual Drive: Immersive VR-Controlled Car with Real-Time Interaction and Live Video Feed." In *2024 International Conference on Computational Intelligence and Computing Applications (ICCICA)*, vol. 1, pp. 519-525. IEEE, 2024. Doi: 10.1109/ICCICA60014.2024.10585050.
4. **Shweta Jain**, Anju Gandhi Bhandari, Devendra Prasad, Stuti Mehla, "Quantum Chain of Things: Amalgamation of Blockchain (Bc), Internet of Things (Iot), And Quantum Computing (Qc)- Qcot Era", *Futuristic Trends in Information Technology Volume 3 Book 1, IIP Series*, Volume 3, May, 2024, Page no.171-192, e-ISBN: 978-93-6252-902-2, Doi: 10.58532/V3BBIT1P2CH2
5. **S. Jain**, R. Aggarwal and A. B. Gandhi, "Quantum chain of things: 'the quantum triad - integrating blockchain, IoT and quantum computing for a transcendent future'," *International Conference on Computer Vision and Internet of Things 2023 (ICCVIoT'23)*, Coimbatore, India, 2023, pp. 299-304, doi: 10.1049/icp.2023.2892.
6. **S. Jain**, A. Gandhi, S. Singla, L. Garg and S. Mehla, "Quantum Machine Learning and Quantum Communication Networks: The 2030s and the Future," *2022 International Conference on Computational Modelling, Simulation and Optimization (ICCMSO)*, Pathum Thani, Thailand, 2022, pp. 59-66, doi: 10.1109/ICCMSO58359.2022.00025.

EMPLOYMENT

Machine Learning Specialist <i>SkillifyMe: Gateway to Tech Mastery, Karnal, Haryana</i>	July 2024 – Aug.2025
Ambassador & AI Researcher <i>All India Council for Technical Education – AICTE IDEA Lab Ambassador</i> <i>Panipat Institute of Engineering & Technology, Haryana</i>	Jan. 2023 – June 2024
Machine Learning Intern <i>Alpine Technologies, Kanpur, Uttar Pradesh ,208011</i>	July 2023 – Aug. 2023
Project Intern <i>Uno Minda Limited, Sonipat, Haryana, 131029</i>	July 2022 – Aug. 2022

PROFESSIONAL SERVICES

Innovation Ambassador, Ministry of Education's Innovation Cell, Government of India	2025 - Present
Research Ambassador, Scifiniti, Abu Dhabi, UAE	2025- Present
Organizing Committee Member (TIACOMP 2024), Bali, Indonesia	2024
PIET ACM Student Chapter Representative	2022-2023

HONORS & ACADEMIC ACHIEVEMENTS

1. GEN AI ACADEMY by Google Cloud and Hack2skill, **Top Performer in the Gen AI Exchange Program 2025**
2. **Qualified University Grants Commission-National Eligibility Test 2025** in Computer Science and Applications.
3. **Rank 1 Holder** in Master of Technology among all the branches in the first year, **National Institute of Technology Delhi, India.**
4. **Best Machine Learning Expert Trainer Recognition by AICTE Sponsored MODROB- "Implementation of Data Analytics Lab**, for delivering trainings to the undergraduate students in the domain of AI, Machine Learning & Data Science.
5. **Best Research Paper Award at ICCICA 2024**, presented my research work at the ICCICA technical co-sponsorship by the IEEE Delhi Section, and academic partnership with AICTE and got best research paper

award by **Prof. (Dr.) Sunil Luthra, Director, Training & Bureau, AICTE (Ministry of Education)**.

6. **Kurukshetra University Gold Medalist** in Bachelor of Technology in AI & ML in all eight semesters, Receiving scholarships and awards for Excellence in academics.
7. **Best Research Paper Award at ICCIMA 2023**, Presented my research work at the ICCIMA, Sponsored by AICTE and publishing partner Taylor and Francis; CRC Journal and got best paper award by Dr. R.K. Soni (Advisor-II, AICTE).

RESEARCH PROJECTS

1. **Quantum – Inspired Grey Wolf Optimization Technique for Intelligent Two -Echelon Electric Vehicle Routing Problems**

Quantum Research and Centre of Excellence, New Delhi

2025

Advisors: Dr. Kumar Gautam, Prof. (Dr) Ajay K. Sharma, and Dr. Divya Punia

Engineered a state-of-the-art Quantum-Inspired Grey Wolf Optimization (Q-GWO) framework to tackle the intricate challenges of Two-Echelon Electric Vehicle Routing Problems (2E-EVRP). This pioneering hybrid quantum-classical approach optimizes multi-depot and intermediate satellite-based logistics networks, seamlessly managing complex constraints such as capacity, energy consumption, and time windows. By leveraging advanced quantum probabilistic modeling, the framework achieves unparalleled routing efficiency, scalability, and sustainability, decisively surpassing traditional optimization techniques and setting a new benchmark for intelligent green logistics.

2. **Deep Reinforcement Learning for Intelligent Drone Delivery Optimization**

National Institute of Technology, Delhi

2025

Advisors: Dr. Divya Punia, Dr. Kumar Gautam, and Prof. (Dr) Ajay K. Sharma

Currently developing an advanced Deep Reinforcement Learning (DRL) framework for intelligent drone delivery optimization. This ongoing research focuses on dynamic, real-time route planning which aims to revolutionize last-mile logistics by enhancing efficiency, scalability, and sustainability while surpassing conventional routing approaches.

3. **Deep Transformer Based Techniques for ENSO & IOD Forecasting**

National Institute of Technology, Delhi

2024-2025

Advisor: Dr. Prashant Kumar

Currently leading an ongoing research project on Deep Transformer-Based Framework for ENSO and IOD Forecasting, focused on developing a GPU-optimized, hybrid Transformer model for long-term, multi-step prediction of complex ocean-atmospheric phenomena. Leveraging high-resolution climate datasets (SST, OHC, SLP, WS) and advanced multi-task learning with Optuna-driven hyperparameter tuning, the framework aims to enhance the accuracy and reliability of Niño 3.4 and DMI forecasts up to 16 -18 months ahead, supporting global climate resilience and early warning systems.

4. **Conversational Summarization & Response Generation using Transformer-based Language Models**

SkillifyMe, Haryana

2024-2025

Collaborator: Mr. Rishit Aggarwal, Indian Institute of Technology Patna

Developed a two-stage framework that combines BART-large for abstractive summarization with Distilled GPT-2 framework for context-aware response generation. This approach significantly improves dialogue coherence and contextual accuracy, making it highly suitable for real-world applications in healthcare, personalized education, and mental health support.

5. **Virtual Drive: Immersive VR-Controlled Car with Real-Time Interaction and Live Video Feed**

Bachelor's Major Thesis Project, AICTE IDEA Lab

Panipat Institute of Engineering & Technology

2024

Advisors: Prof. (Dr.) Anju Gandhi & Prof. (Dr.) Sunil Dhull – AICTE IDEA Lab Heads

Designed and developed an innovative VR-controlled car system enabling real-time interaction and remote navigation through a live video feed. The project integrated virtual reality, IoT sensors, and

computer vision to create an immersive driving experience, enhancing remote vehicle operation and showcasing next-generation human-vehicle interaction for autonomous systems and smart transportation.

6. **Technological Marvels: Pioneering PCOS Detection through the Lens of Machine Learning and Deep Learning**

Bachelor's Minor Project, AICTE IDEA Lab
Panipat Institute of Engineering & Technology
Advisor: Prof. (Dr.) Anju Gandhi

2023

From detection to Diagnosis: Developed a machine learning and deep learning-based diagnostic system to accurately detect Polycystic Ovary Syndrome (PCOS). Leveraging healthcare datasets, the project applied advanced algorithms for early detection and personalized health insights, supporting timely intervention and improved women's healthcare outcomes.

7. **Smart Health and Disease Prediction System**

Panipat Institute of Engineering & Technology
Advisor: Prof. (Dr.) Anju Gandhi

2022

Built an intelligent platform capable of predicting various diseases using patient health data and symptoms. The system utilized predictive analytics and AI-driven models to assist doctors and patients in early diagnosis and preventive healthcare planning.

WORKSHOP & INVITED TALKS

1. Delivered a 5 days' workshop on **National Technology Day 2024** titled "**Research Projects in the field of AI**", Sponsored by **DST, Haryana**, and supported by Inner Wheel Central Panipat, organized by the Dept. of CSE AIML, Panipat Institute of Engineering & Technology. Guided 100+ students and delivered expert talks to undergraduate AIML students on research-based projects, research publications.
2. Invited as an AI Expert Talk & Judge at **PRASHIKSHAN 2025** Organized by Department of Computer Science & Engineering, Panipat Institute of Engineering & Technology, Panipat. Evaluating innovative research projects and providing strategic guidance to undergraduate students.
3. Delivered a **Career Guidance Session Expert Talk 2025** in Department of Computer Science & Engineering, PIET. Mentoring first- and second-year undergraduates on career paths and opportunities in AI research and emerging technologies.

REVIEWERS

1. **Recognition by Dr. Alessio Giorgetti, IEEE Access Associate Editor (2024)**: Got Recognition by IEEE Access as an IEEE Reviewer for reviewing the research manuscript "A Survey on Quantum Machine Learning: Basics, Current Trends, Challenges, Opportunities, and the Road Ahead" by New York University.
2. Serving as a **Peer Reviewer** for four manuscripts currently under consideration for publication in the **edited volume titled "Neural Network Technologies and Brain-Computer Interfaces: Innovations and Applications"** by IGI Global Scientific Publishing. The manuscripts under my review include Introduction of a Cutting-Edge Technology Neural Networks Organization, Bridging Neural Technologies with Financial Decision-Making in Neurofinance, Dynamic Rehabilitation Through Brain-Computer Interfaces Integrating CNN and Bi-GRU Deep Learning Approach, Deep Learning for BCI Signal Decoding

UNDERGRADUATE SUMMER RESEARCH PROJECT SUPERVISION

Guided and mentored 250+ undergraduate students through research-based summer projects, providing hands-on training in AI, Machine Learning, and advanced computational techniques. Supervised students in developing innovative solutions – based projects, and enhancing their research methodology.

CERTIFICATIONS

1. **Google Generative AI Academy by Google Cloud and Hack2skill 2025:** Generative AI, exploring tools and technologies like: Vertex AI, Gemini APIs, Imagen, Streamlit and Multimodal RAG
2. **Introduction to Quantum Information by Korea Advanced Institute of Science and Technology**

OTHER POSITION OF RESPONSIBILITY & ACTIVITIES

1. **Training & Placement Cell Coordinator, National Institute of Technology, Delhi, 2025** Coordinating campus recruitment drives, industry outreach, and student career development initiatives, fostering strong collaborations between students and leading organizations.
2. Conference Host & Organizing Committee Member at **International Conference on Progressive Innovations in Intelligent Systems and Data Science (ICPIDS 2024), Pattaya, Thailand:** Collaborated with International Scientific Research Foundation and Novel Research Foundation, to organize and coordinate the event, which brought together experts and scholars in the field of advanced computing.
3. Conference Host **International Conference on Technological Innovations and Advanced Computing (TIACOMP 2024), Bali, Indonesia:** Collaborated with Novel Research Foundations, India, to organize and coordinate the event, which brought together experts and scholars in the field of advanced computing.
4. Mathematics & Computing Branch Representative (2024 – Present) Serving as the elected representative for the **Mathematics & Computing Master's Program**, facilitating communication between faculty and students, addressing concerns, and driving academic and research-focused initiatives.
5. **Overall Student Head & Book Editor ,1st International Conference on Computational Intelligence and Computing Applications (2024):** Directed and coordinated all conference activities, guided team members, guests, and keynote speakers, and managed data, publishing, book compilation, IEEE proceedings, and abstract book preparation.
6. **Student Coordinator Head, 1st International Conference on Computational Intelligence and Mathematical Applications (2023):** Organized a 2-day international conference in Panipat Institute of Engineering of Technology. Managed all the proper scheduling of the event, Paper publishing in CRC, transportation, accommodation, and educational-related activities.
7. Presented the **research paper** titled "Integrating Quantum Concepts with Machine Learning " at 1st International Conference on Futuristic Computation Techniques: Approaches, Implementations and Applications (ICFCT -2022).
8. **Artificial Intelligence and Machine Learning Representative (2022-2024):** Elected as the Major Class Representative for AIML, entrusted with the responsibility to bridge the gap between students and faculty, fostering a collaborative and inclusive learning environment.
9. **Student Coordinator, International's Women Day, PIET (2021):** Orchestrated and coordinated activities and events in celebration of International Women's Day at PIET. Managed logistical aspects, including event planning, coordination, and resource allocation.
10. **Event Coordinator and Anchor, National Engineers Day (2021):** Managed and hosted the event; Guided students for the projects also and got appreciation from BOG, and PIET regarding the best coordination and management.

REFERENCES

1. Dr. Prashant Kumar (Advisor in MTech):

Associate Professor, Department of Applied Sciences

National Institute of Technology Delhi

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Email: prashantkumar@nitdelhi.ac.in

Web Page: <https://nitdelhi.irins.org/profile/62655>

2. Prof. (Dr.) Ajay K Sharma, Director (Co-Advisor in MTech):

National Institute of Technology Delhi

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Tele: +911133861001(O); 8588833111-222

Email: director@nitdelhi.ac.in / sharmanitd@gmail.com

Website: www.ajayksharma.com

3. Prof. (Dr.) Anju Gandhi (Advisor in BTech):

Professor – CSE, Professor Incharge(R&D), AICTE IDEA LAB HEAD and IIC HEAD

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